

Rentify — Technical & Design Specification

Version: 1.0 (MVP) **Companion to:** Rentify PRD v1.0

1. Tech Stack

Layer	Choice	Why
Frontend (Web)	React.js + Tailwind CSS	Fast to build, mobile-responsive, huge community support for a solo/small team
Mobile (later phase)	React Native	Reuse most of the web logic/components when you go native
Backend	Node.js + Express.js	JavaScript everywhere = one language across the whole stack, easier for a small team
Database	PostgreSQL	Structured data (users, listings, transactions) with relations — fits this domain well
Real-time Chat	Firebase Realtime Database or Socket.io	Instant messaging without building infra from scratch
Auth	Firebase Authentication (Google + Email/Phone OTP)	Handles verification, OTP, and Google login out of the box
File/Image Storage	Firebase Storage or AWS S3	For listing photos and profile pictures
Hosting (Frontend)	Vercel	Free tier, zero-config React deployment
Hosting (Backend)	Render or Railway	Low-cost, simple deploys for a Node/Express API in MVP stage
Notifications	Firebase Cloud Messaging (FCM)	Push notifications for chat/deal updates
Analytics	Google Analytics / Mixpanel (basic)	Track which features are actually used before scaling

2. Page-by-Page Workflow

2.1 Onboarding

Splash Screen

↓
Login (Google / College Email / Phone OTP)
↓
Profile Setup (Name, Branch, Year, Photo) – ~2 min
↓
Landing Page

2.2 Landing Page

[Top Bar: Logo | Search | Notification Bell | Profile Icon]
[Category Chips: Books | Electronics | Furniture | Project Parts | Misc]
[Two Primary Buttons: "Browse Items" | "Post an Item"]
[Feed: Personalized listings based on Branch/Year]

2.3 Post an Item Flow

Tap "Post an Item"
↓
Choose: Rent OR Sell
↓
Select Category
↓
Enter Title, Description, Condition
↓
Upload Photos (min. 1)
↓
Set Price (+ duration if Rent)
↓
Set Pickup Location
↓
Review & Publish
↓
Listing goes live as "Available"

2.4 Browse & Discover Flow

Tap "Browse Items"
↓
Search bar + Filters (Category / Price / Branch-Year / Rent-Sale / Distance)
↓
Listing Grid/List View
↓
Tap a Listing → Item Detail Page
(Photos, Price, Condition, Seller Card with rating & verified badge)
↓
Tap "Message Seller"

2.5 Chat & Deal Flow

Chat Thread Opens
↓
Quick-reply templates OR free typing
↓
Seller marks listing as "Reserved" (optional, once serious)
↓
Both parties meet & exchange item
↓
Both tap "Deal Completed" independently
↓
Listing status → Sold / Rented Out
↓

Rating prompt triggers for both users

2.6 Profile & Listings Management

My Profile

- └─ My Listings (tabs: Active / Reserved / Completed)
 - └─ Ratings & Reviews Received
 - └─ Saved/Wishlist Items
 - └─ Account Settings
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3. APIs Required

3.1 Authentication

- POST /auth/google — Google OAuth login
- POST /auth/phone/send-otp — Send OTP to phone
- POST /auth/phone/verify-otp — Verify OTP and log in
- POST /auth/email/verify — College email verification (for “Verified” badge)

3.2 User & Profile

- GET /users/:id — Fetch profile
- PUT /users/:id — Update profile (branch, year, photo)
- GET /users/:id/ratings — Get all ratings received

3.3 Listings

- POST /listings — Create a new listing
- GET /listings — Browse/search listings (supports query params: category, price range, branch, rent/sale, location)
- GET /listings/:id — Get single listing detail
- PUT /listings/:id — Edit a listing
- PATCH /listings/:id/status — Update status (Available / Reserved / Sold / Rented Out)
- DELETE /listings/:id — Remove a listing

3.4 Chat

- POST /chats — Start a new chat thread (tied to a listing + buyer + seller)
- GET /chats/:id/messages — Fetch messages in a thread
- POST /chats/:id/messages — Send a message
- WS /chats/:id — WebSocket/real-time connection for live messages

3.5 Deals & Ratings

- POST /deals/:listingId/confirm — Buyer or seller confirms deal completion
- POST /ratings — Submit a rating (triggered only after both-side deal confirmation)
- GET /ratings/:userId — Fetch a user’s rating summary

3.6 Notifications

- POST /notifications/register-device — Register device token for push (FCM)
- GET /notifications — Fetch a user’s notification list

3.7 Safety

- POST /reports — Report a user or listing
- POST /users/:id/block — Block a user

3.8 Image Upload

- POST /uploads/image — Upload a listing/profile photo, returns a storage URL
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4. Design System

4.1 Design Direction

Rentify's world is campus life — practical, fast-paced, budget-conscious, but still needs to feel trustworthy enough that a stranger will hand over their money or their book. The design leans into that: warm and approachable rather than corporate, with a clear visual language for “available,” “reserved,” and “gone” so a student can scan a feed in seconds between classes.

4.2 Color Palette

Name	Hex	Usage
Notebook Cream	#FAF6EE	Primary background — warm, paper-like, easy on the eyes for long browsing sessions
Ink Navy	#1F2A44	Primary text, headers — feels trustworthy and legible, not harsh black
Marigold	#E8A33D	Primary accent — CTAs like “Post an Item,” active states (chosen over the common terracotta/orange-AI-default; marigold has a more campus/stationery feel, evokes highlighter pens and campus event flyers)
Moss Green	#4C7A5E	Success states — “Available,” “Deal Completed,” verified badge
Rust Clay	#B5533C	Warning/status — “Reserved” tag, gentle urgency without alarm
Slate Grey	#7C8595	Secondary text, borders, disabled states

Avoid pure black (#000) and pure white (#FFF) — use Ink Navy and Notebook Cream as the anchors instead for a warmer, less clinical feel.

4.3 Typography

Role	Typeface	Notes
Headlines	Fraunces (serif)	Has personality and warmth — distinct from the generic sans-serif SaaS look, fits a “campus notice board” feel
Body & UI text	Inter (sans-serif)	Highly legible at small sizes for listings, chat, forms

- Line length: keep body text under 75 characters per line for readability
- Headline weight: Semibold/Bold for section titles; avoid all-caps labels
- Use size and weight — not color — to establish hierarchy where possible

4.4 Layout Principles

- **Card-based listing grid** on Browse, but avoid uniform rounded-corner cards with identical shadows on everything — vary emphasis: the item photo is the hero of each card, price and status badge sit close to it, seller info stays secondary and small
- **Left-aligned text** throughout (not centered) — this is a utility app, not a marketing page; center alignment is reserved only for the empty-state screens
- **Status badges** (Available/Reserved/Sold) use color + a short word, never color alone, so it stays accessible
- **Bottom navigation bar** on mobile: Home, Browse, Post (raised/prominent center button), Chats, Profile

4.5 Interaction Notes

- One meaningful animation: a subtle upward reveal when a new listing is successfully posted (confirmation moment), not scattered hover effects on every card
- “Mark Deal Complete” button changes state visibly and immediately (shows “Waiting for other party” until both confirm) — clarity over decoration
- Empty states (e.g., no listings in a filter) should suggest a next action (“Try a different category” or “Be the first to post here”) rather than just showing “No results”